



SOFTWARE REQUIREMENTS SPECIFICATION

NutriVision

Abstract

This document defines the functional and non-functional requirements for the NutriVision web application. It serves as the primary reference for understanding the system's features, expected behavior, and quality requirements, supporting the planning and execution of software testing activities.

Prepared By
Alishba Rashid

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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

NutriVision – Smart Food Recognition & Nutrition Analyzer

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Prepared By: Alishba Rashid

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Revision History

Version	Date	Author	Description
1.0	30 June 2026	Alishba Rashid	Initial version

Introduction

Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of **NutriVision – Smart Food Recognition & Nutrition Analyzer**. The document serves as the foundation for planning, designing, and executing manual software testing while providing a clear understanding of the application's intended functionality.

Scope

NutriVision is a responsive web application that enables users to analyze food images, manually search food items, monitor daily nutritional intake, and manage personal health information. The application simulates AI-powered food recognition using mock services and provides users with nutritional insights through an intuitive and modern interface.

Intended Audience

This document is intended for:

- Software Developers
- QA Engineers
- Project Reviewers

- Recruiters and Hiring Managers

System Overview

NutriVision is a web-based application developed using **React**, **TypeScript**, and **Tailwind CSS**. It provides users with an interactive platform to analyze food images, search nutritional information, monitor dietary progress, and manage personal fitness goals. The application uses mock AI services to simulate food recognition and nutrition analysis while maintaining an architecture that supports future backend integration.

Functional Requirements

Landing Page (FR-1)

The system shall:

- **FR-1.1** Display a hero section introducing the application.
- **FR-1.2** Display feature highlights.
- **FR-1.3** Display a "How It Works" section.
- **FR-1.4** Display customer testimonials.
- **FR-1.5** Display Frequently Asked Questions (FAQ).
- **FR-1.6** Display a contact form.
- **FR-1.7** Validate contact form inputs.
- **FR-1.8** Redirect users to the Authentication page when the **Get Started** button is clicked.

Authentication (FR-2)

The system shall:

- **FR-2.1** Allow users to create a new account.
- **FR-2.2** Allow registered users to sign in.
- **FR-2.3** Allow users to recover forgotten passwords.
- **FR-2.4** Validate email format.
- **FR-2.5** Enforce a minimum password length of six characters.
- **FR-2.6** Prevent unauthorized access to protected pages.
- **FR-2.7** Maintain authenticated user sessions.
- **FR-2.8** Allow users to log out successfully.

Dashboard (FR-3)

The system shall:

- **FR-3.1** Display a personalized welcome message.

- **FR-3.2** Display daily calorie progress.
- **FR-3.3** Display protein progress.
- **FR-3.4** Display carbohydrate progress.
- **FR-3.5** Display fat progress.
- **FR-3.6** Display a water intake tracker.
- **FR-3.7** Display weekly calorie analytics.
- **FR-3.8** Display recently analyzed food items.

AI Food Analyzer (FR-4)

The system shall:

- **FR-4.1** Allow users to upload food images using drag-and-drop or file selection.
- **FR-4.2** Accept JPEG and PNG image formats.
- **FR-4.3** Reject unsupported file formats.
- **FR-4.4** Display a preview of the uploaded image.
- **FR-4.5** Allow users to replace or remove the selected image.
- **FR-4.6** Simulate AI-based food analysis.
- **FR-4.7** Display a loading indicator during analysis.

Analysis Results (FR-5)

The system shall:

- **FR-5.1** Display the uploaded food image.
- **FR-5.2** Display the food health status.
- **FR-5.3** Display the overall health score.
- **FR-5.4** Display calorie information.
- **FR-5.5** Display protein content.
- **FR-5.6** Display carbohydrate content.
- **FR-5.7** Display fat content.
- **FR-5.8** Display fiber content.
- **FR-5.9** Display sugar content.
- **FR-5.10** Display AI-generated nutritional insights.
- **FR-5.11** Display healthier food recommendations.

Manual Food Search (FR-6)

The system shall:

- **FR-6.1** Allow users to search food items manually.
- **FR-6.2** Display matching food suggestions.
- **FR-6.3** Display nutritional information for the selected food item.
- **FR-6.4** Allow users to add the selected food item to the daily nutrition log.

History (FR-7)

The system shall:

- **FR-7.1** Display previously analyzed food records.
- **FR-7.2** Allow users to search history records.
- **FR-7.3** Allow users to filter history records.
- **FR-7.4** Allow users to delete individual history records.
- **FR-7.5** Allow users to clear all history after confirmation.

Profile & Settings (FR-8)

The system shall:

- **FR-8.1** Allow users to edit personal profile information.
- **FR-8.2** Allow users to update fitness goals.
- **FR-8.3** Support Dark Mode.
- **FR-8.4** Allow users to manage notification preferences.
- **FR-8.5** Allow users to change language preferences.
- **FR-8.6** Allow users to configure privacy settings.

Non-Functional Requirements

The application shall:

- Be fully responsive across desktop, tablet, and mobile devices.
- Provide a modern, intuitive, and user-friendly interface.
- Support smooth animations and transitions.
- Validate user input on the client side.
- Use TypeScript to ensure type safety.
- Maintain a modular and scalable code structure.
- Simulate asynchronous API responses using mock services.
- Deliver consistent performance under normal usage conditions.

Assumptions & Limitations

- AI food recognition and nutritional analysis are simulated using mock services.
- Nutritional values are intended for demonstration and testing purposes only.
- Internet access is required to use the application.
- The application is designed for learning, demonstration, and manual QA testing.